LUMPI: A LiDAR Object-Detection Research Prototype Using Height Filtering, Voxel Downsampling, and Density-Based Clustering

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***Abstract*—Three-dimensional light detection and ranging (LiDAR) provides geometric measurements that can support object detection in autonomous systems and environmental perception. This paper presents LUMPI, a preliminary LiDAR object-detection research prototype based on height filtering, voxel downsampling, density-based spatial clustering of applications with noise (DBSCAN), cluster-derived candidate boxes, and approximate three-dimensional evaluation. The reported experiment processes 31 scans from a single measurement subset. It generates 28,242 candidate proposals and evaluates them against 1,792 provisionally selected ground-truth boxes. At an axis-aligned three-dimensional intersection-over-union (IoU) threshold of 0.25, the system obtains a micro-precision of 0.0364, a recall of 0.5737, and a micro-F1 score of 0.0685. At IoU 0.50, micro-precision is 0.0089, recall is 0.1401, and micro-F1 is 0.0167. These results show that the prototype retrieves a substantial fraction of the provisionally selected annotations at the more permissive threshold, but produces many unmatched proposals and performs poorly under the stricter threshold. The experiment is preliminary: the annotation-score filter, scan-to-label temporal alignment, class semantics, and orientation-aware evaluation have not been independently verified against the dataset's official specification. Consequently, the reported numbers are internal diagnostic measurements, not official benchmark scores. This work documents a reproducible baseline and identifies the validation steps required before drawing stronger conclusions about detection performance or research novelty.**

***Index Terms*—LiDAR, point clouds, object detection, DBSCAN, voxel downsampling, three-dimensional perception, evaluation methodology**

I. Introduction

LiDAR sensors represent the surrounding environment as a set of three-dimensional points. These measurements can support geometric perception tasks such as obstacle detection, scene reconstruction, and object localization. Unlike image-only methods, point-cloud methods directly process spatial measurements, although the points may be sparse, noisy, and unevenly distributed.

Three-dimensional object detection remains challenging because different objects can produce different point densities depending on distance, orientation, occlusion, and sensor configuration. Nearby objects may merge into a single connected region, while a single object may be fragmented into multiple clusters. These effects are particularly important for methods that infer objects from geometric connectivity.

LUMPI is a preliminary research prototype intended to establish a transparent, measurable baseline using conventional point-cloud operations. The pipeline applies a height filter, downsamples points using a voxel grid, groups spatially neighboring points with DBSCAN, and derives candidate bounding boxes from the resulting clusters. Candidate boxes are then compared with annotations using a simplified matching and evaluation procedure.

The primary contribution is an experimental record of this baseline, including proposal counts, aggregate detection metrics, per-scan variation, and visual diagnostics. The paper does not claim that the pipeline establishes a new detection algorithm. Instead, it identifies what the current implementation measures, where its evaluation is approximate, and what must be verified before comparing it with established methods.

The main research questions are:

1) Can a simple clustering pipeline generate candidate object regions from the available LiDAR scans?
2) How do its precision and recall change when the matching IoU threshold increases?
3) What failure modes are suggested by the proposal counts and per-scan evaluation results?
4) Which methodological uncertainties must be resolved before the results can support a defensible benchmark claim?

II. Related Work

A. Density-Based Clustering

DBSCAN is a density-based clustering algorithm introduced by Ester et al. [1]. It identifies dense regions using a neighborhood radius and a minimum number of neighboring samples. Unlike algorithms that require the number of clusters to be specified in advance, DBSCAN can identify multiple clusters and label sparse points as noise.

This property makes DBSCAN a useful baseline for point-cloud exploration. However, density connectivity is not equivalent to semantic object identity. A single physical object may fragment into several clusters, while neighboring objects may be merged into one cluster. Parameter choices also affect the result.

B. LiDAR Benchmarks

The KITTI benchmark introduced widely used evaluation tasks for autonomous-driving perception, including three-dimensional object detection [2]. Benchmark evaluation depends on consistent annotation definitions, coordinate conven-

TABLE I
PROVISIONAL ANNOTATION FIELD INTERPRETATION

Index	Provisional meaning
0	Time or frame-related value
1	Object identifier
2–5	Image-space box coordinates and dimensions
6	Score or annotation status; semantics unverified
7	Class identifier; mapping unverified
8	Visibility-related field; semantics unverified
9–11	Three-dimensional center coordinates
12–14	Three-dimensional dimensions
15	Heading or orientation
16	Additional field; semantics unverified

tions, object categories, difficulty conditions, and matching rules. A simplified axis-aligned overlap metric should therefore not be treated as interchangeable with an official oriented 3D evaluation protocol.

C. Point-Cloud Processing

Point-cloud libraries provide operations for filtering, geometric processing, and voxel downsampling. Voxel downsampling can reduce the number of points by representing local regions with fewer samples, lowering computational requirements at the cost of geometric detail [4].

D. Learned Three-Dimensional Detection

PointPillars converts point clouds into vertical pillars and applies a learned detection architecture to estimate object classes and bounding boxes [5]. It is an example of a substantially more developed detection approach than the geometry-only LUMPI baseline. A fair comparison would require the same dataset split, annotation conventions, evaluation protocol, and clearly specified training and inference settings.

The present work uses these established methods as context, not as evidence that LUMPI outperforms them. No direct benchmark comparison is claimed.

III. DATASET AND EXPERIMENTAL SCOPE

A. Subset Selection

The reported experiment uses the Measurement5 subset. Thirty-one scans were processed, with no reported processing failures. The annotation file was reported to contain 2,082 rows and 17 positional columns. After applying the provisional annotation filter, 1,792 ground-truth boxes were used in the evaluation.

The experiment generated 28,242 candidate proposals in total. These counts describe the processed subset only and should not be interpreted as dataset-wide performance.

B. Annotation Interpretation

The annotation file contains repeated or ambiguous column names. The implementation therefore accesses fields by their positional indices. The following positional interpretation was used:

The filter `score == 4` was used provisionally to select ground-truth annotations. This value has not been verified against an authoritative dataset specification. The resulting ground-truth count and all metrics that depend on it must therefore be treated as conditional on this assumption.

Likewise, the class identifiers have not been authoritatively confirmed. No claim about per-class accuracy is made in this paper.

C. Temporal Alignment

The experiment associated scan index i with a timestamp $i/10$, corresponding to a nominal interval of 0.1 seconds. This numerical mapping was used in the initial evaluation but has not been independently verified as the official scan-to-label association.

A timestamp sequence that appears numerically consistent does not by itself establish that each point cloud and annotation row describe the same instant. Temporal alignment must be checked against the dataset metadata before the evaluation can be regarded as definitive.

IV. METHODOLOGY

A. Pipeline Overview

The baseline consists of the following stages:

1) Load a LiDAR point cloud.
2) Apply a height-based point filter.
3) Downsample the filtered cloud using a voxel grid.
4) Apply DBSCAN to group spatially dense points.
5) Derive a candidate bounding box for each retained cluster.
6) Match candidates to ground-truth boxes using IoU.
7) Aggregate true-positive, false-positive, and false-negative counts across scans.

The pipeline is intentionally simple. Its purpose is to establish a baseline and expose failure modes, rather than to represent a state-of-the-art learned detector.

B. Height Filtering

Let a point be

$$\mathbf{p}_i = (x_i, y_i, z_i).$$

The height-filtering stage retains points satisfying

$$z_i \geq z_{\min},$$

where the reported setting is

$$z_{\min} = -1.8.$$

This threshold is expressed in the input coordinate system. Its physical interpretation depends on the coordinate origin, axis convention, and units. These must be confirmed from the dataset documentation.

Height filtering reduces some unwanted points, but it can also remove useful object points or retain ground and background structures when the coordinate convention is misunderstood.

C. Voxel Downsampling

The filtered points are downsampled using a voxel size of

$$v = 0.12.$$

Voxel downsampling reduces the number of points that the clustering stage must process. Its benefit is lower computational cost; its limitation is the loss of fine geometric detail.

The selected voxel size is an experimental setting, not an established optimum. It should be evaluated through controlled parameter sweeps using a fixed validation protocol.

D. DBSCAN Clustering

DBSCAN identifies density-connected groups using a neighborhood radius ϵ and a minimum sample count m.

The reported settings are

$$\epsilon = 0.65, \qquad m = 6.$$

Points that do not belong to a sufficiently dense region may be classified as noise. Each retained cluster is treated as a candidate object region.

These parameters affect cluster fragmentation and merging. A larger radius may join neighboring objects, while a smaller radius may split a single object into multiple clusters. The best values depend on point density, distance, sensor properties, and the intended object classes.

E. Candidate Bounding Boxes

For each retained cluster, the baseline derives a candidate box from the cluster's spatial extent. This provides a straightforward way to compare clustered regions with annotated boxes.

However, a cluster-derived box is not guaranteed to correspond to a single physical object. It may include background points, omit parts of an object, or combine several objects. Such geometric errors directly affect the overlap-based evaluation.

The baseline is therefore best understood as a candidate generation method, not as a complete semantic detector.

F. Intersection over Union

For two three-dimensional boxes A and B, the intersection over union is

$$\mathrm{IoU}(A, B) = \frac{\mathrm{Vol}(A \cap B)}{\mathrm{Vol}(A) + \mathrm{Vol}(B) - \mathrm{Vol}(A \cap B)}.$$

The reported implementation uses an approximate axis-aligned three-dimensional overlap calculation. It does not fully account for box heading. When ground-truth objects are rotated, this approximation can differ substantially from an oriented-box calculation.

The evaluation uses IoU thresholds of 0.25 and 0.50. Results at these thresholds are diagnostic and must not be interpreted as official benchmark scores.

TABLE II
REPORTED BASELINE CONFIGURATION

Setting	Value
Dataset subset	Measurement5
Processed scans	31
Height threshold	-1.8
Voxel size	0.12
DBSCAN radius	0.65
DBSCAN minimum samples	6
Provisional GT filter	`score == 4`
Timestamp mapping	index / 10
Overlap thresholds	0.25, 0.50
Overlap geometry	Approximate, axis-aligned 3D
Matching method	Greedy one-to-one

TABLE III
REPORTED AGGREGATE DETECTION RESULTS

Metric	IoU 0.25	IoU 0.50
TP	1,028	251
FP	27,214	27,991
FN	764	1,541
Precision	0.0364	0.0089
Recall	0.5737	0.1401
Micro-F1	0.0685	0.0167

G. Matching and Metrics

A greedy one-to-one matching procedure is used to associate predicted boxes with ground-truth boxes. The procedure determines true positives (TP), false positives (FP), and false negatives (FN) for each scan.

The metrics are

$$\mathrm{Precision} = \frac{TP}{TP + FP},$$

$$\mathrm{Recall} = \frac{TP}{TP + FN},$$

and

$$F_1 = 2\frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}.$$

Aggregate micro-metrics are computed from the summed TP, FP, and FN counts. Mean per-scan F1 is computed separately by averaging the scan-level F1 values. These two summaries are not identical in general because they aggregate the data differently.

V. EXPERIMENTAL CONFIGURATION

The values in Table II describe the recorded baseline run. They are not presented as optimal parameters. A reproducible benchmark requires recording software versions, randomness settings where relevant, preprocessing details, and the exact official evaluation procedure.

VI. QUANTITATIVE RESULTS

A. Aggregate Detection Metrics

Table III presents the reported aggregate results for the two IoU thresholds.

At IoU 0.25, the detector achieves recall of approximately 57.4%, but precision is approximately 3.64%. This indicates that many ground-truth boxes receive a match under the permissive overlap threshold, while the vast majority of proposals remain unmatched.

At IoU 0.50, recall drops to approximately 14.0%, and precision drops below 1%. The stricter threshold exposes localization inaccuracies, cluster merging, cluster fragmentation, and other sources of overlap error.

These observations are consistent with a candidate generator that produces many rough spatial regions but lacks effective proposal filtering and accurate object localization. The results alone do not establish which failure mode is dominant.

B. Proposal Counts

Across the 31 scans, the system generated 28,242 proposals. The reported mean was 911.03 proposals per scan, with a standard deviation of 16.88, a minimum of 869, and a maximum of 935.

The relatively narrow range of proposal counts suggests that the pipeline produced a similar number of candidate regions in each processed scan. This does not imply that the proposals were accurate, correctly classified, or spatially consistent.

C. Per-Scan Variation

The reported per-scan F1 at IoU 0.25 had a mean of 0.0685, a standard deviation of 0.0042, a minimum of 0.0605, and a maximum of 0.0775. At IoU 0.50, the mean was 0.0167, the standard deviation was 0.0033, the minimum was 0.0101, and the maximum was 0.0251.

These statistics suggest limited variation in F1 across the selected scans relative to the low aggregate scores. Because all scans come from a single subset and temporal independence has not been established, this is not evidence of generalization to different scenes or environments.

D. Classification Versus Detection

A separate exploratory experiment reportedly trained a classifier using ground-truth boxes and obtained accuracy of 0.992 and macro-F1 of 0.9757. Those numbers describe a classification experiment on annotated boxes, not the complete end-to-end detector.

The classifier was given ground-truth regions rather than having to discover and localize objects from raw point clouds. Its metrics must therefore remain separate from the detection metrics in Table III. They do not demonstrate high end-to-end object-detection performance.

VII. Visual Diagnostics

The figures in this section are intended to help inspect the reported measurements and understand possible failure modes. Their interpretation should be checked against the actual rendered images and their plotting code. Figure captions describe the intended diagnostic purpose; they do not substitute for inspection of the underlying data.

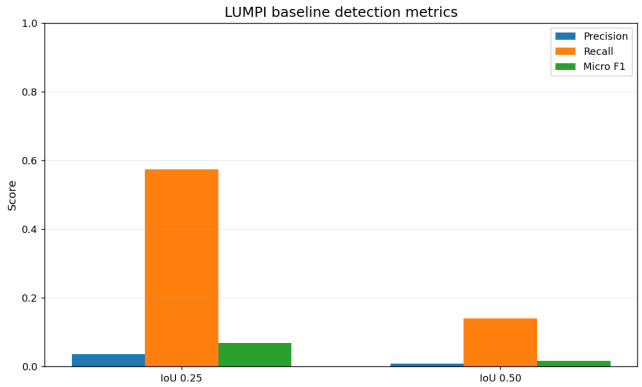


Fig. 1. Aggregate precision, recall, and F1 diagnostics at the evaluated IoU thresholds. The main pattern to inspect is the decline in performance at the stricter overlap threshold.

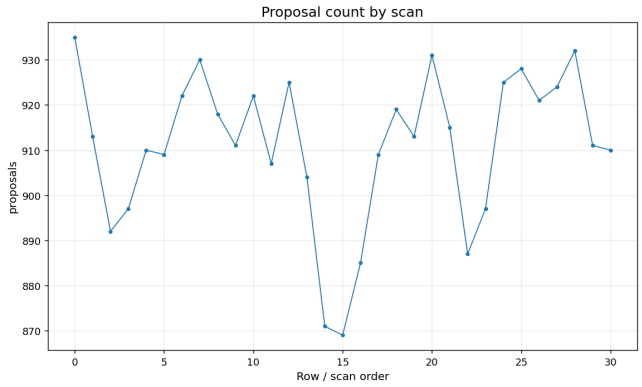


Fig. 2. Number of candidate proposals generated per scan in the Measurement5 experiment.

A. Aggregate Metrics

Figure 1 should be read together with Table III. The permissive IoU threshold allows more approximate overlaps to count as matches, whereas the stricter threshold demands better localization. Any displayed values should be verified against the recorded aggregate metrics.

B. Proposal Counts

Figure 2 provides a view of the number of candidate regions generated by the pipeline. Counts near the reported mean of 911 proposals per scan indicate that proposal generation was numerically consistent across these scans. Proposal volume alone, however, is not a measure of detection quality. A useful next analysis would relate proposal counts to false-positive counts and ground-truth object counts per scan.

C. Per-Scan F1

Figure 3 is intended to expose scan-level variation that aggregate metrics can conceal. It should be checked for outliers, trends, and differences between overlap thresholds. The reported summary statistics indicate low F1 values across the evaluated scans, but they do not establish performance on other subsets.

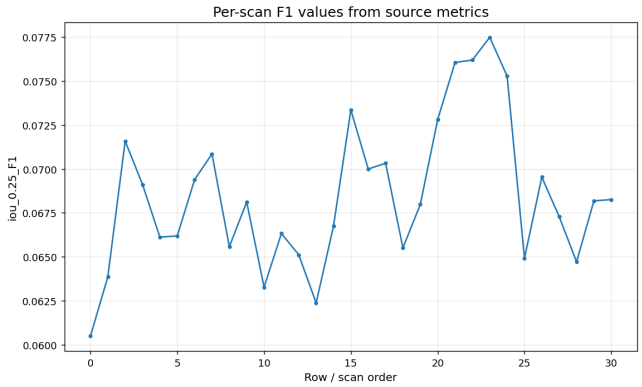


Fig. 3. Per-scan F1 values reported by the evaluation pipeline.

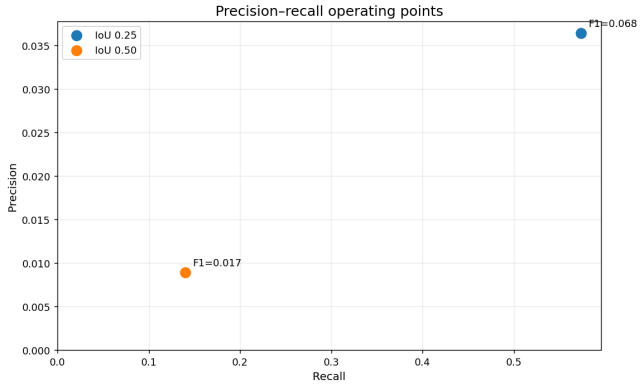


Fig. 4. Precision–recall diagnostic for the LUMPI baseline.

D. Precision–Recall Diagnostic

Figure 4 should be interpreted according to the data and thresholds actually plotted. Two operating points at different IoU thresholds are not, by themselves, a complete precision–recall curve swept over a confidence threshold. If the image presents a full curve, its generation procedure and threshold values should be documented. Otherwise, it should be described as a comparison of operating points rather than a continuous curve.

E. Three-Dimensional Point Cloud

Figure 5 is intended to support visual inspection of the point distribution and coordinate structure. Such a visualization can help reveal parser errors, unexpected coordinate scales, missing regions, or a coordinate orientation that differs from expectations. It cannot, on its own, prove that the annotation timestamps are correctly aligned with the point cloud.

F. Bird's-Eye View and Ground Truth

Figure 6 can be used to inspect the relative placement of the point cloud and the ground-truth annotations. This is especially important before interpreting overlap metrics. A systematic translation, axis swap, sign error, or coordinate unit mismatch could produce poor IoU even when the point-cloud parser itself is functioning correctly.

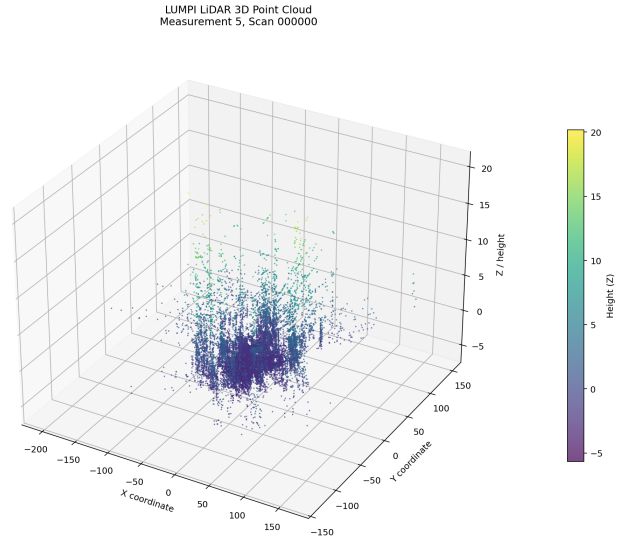


Fig. 5. Example three-dimensional LiDAR point-cloud visualization.

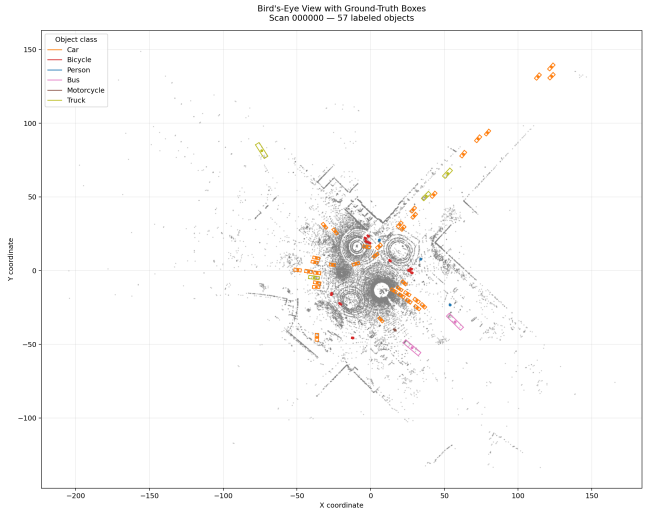


Fig. 6. Bird's-eye view of the LiDAR scene and its ground-truth annotations.

G. DBSCAN Clusters and Ground Truth

Figure 7 is intended to reveal whether clusters correspond approximately to annotated objects. Cluster splitting, cluster merging, background inclusion, and missing object points are plausible failure modes for density-based grouping. Their actual presence and frequency should be established by inspecting the plotted scene and corresponding annotations.

H. Detection Evaluation Visualization

Figure 8 should be used as a diagnostic rather than as evidence of an official benchmark result. Its interpretation depends on the displayed matching procedure, the box representation, and the definitions used for true positives, false positives, and false negatives.

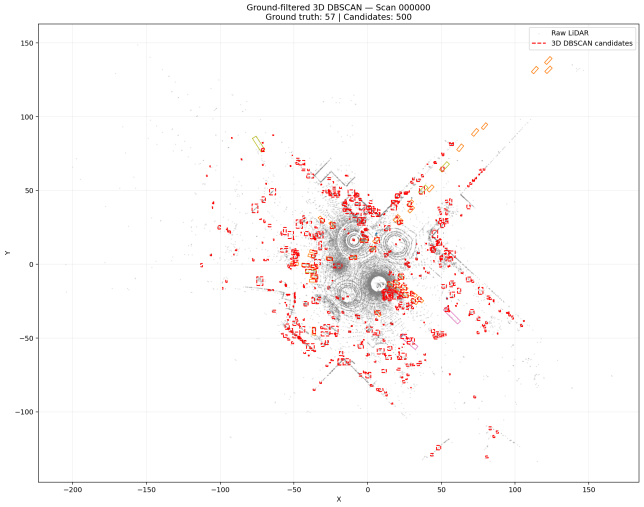


Fig. 7. Three-dimensional diagnostic comparing DBSCAN-derived regions with ground-truth annotations.

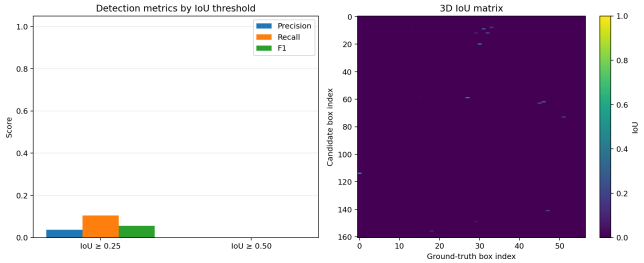


Fig. 8. Visualization of the DBSCAN candidate detections and evaluation results.

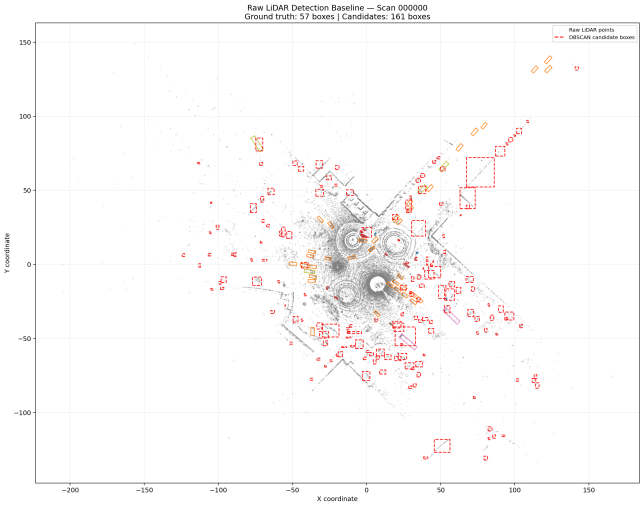


Fig. 9. Visual comparison between candidate predictions and ground-truth boxes.

I. Predictions Versus Ground Truth

Figure 9 can help identify mislocalization, over-sized boxes, under-sized boxes, and candidate boxes that do not correspond to an annotated object. For oriented objects, axis-aligned overlap may conceal important heading errors or overestimate the intersection of rotated boxes. A subsequent evaluation should use a validated oriented-box intersection calculation.



Fig. 10. Detection error-count diagnostic for the evaluated baseline.

J. Detection Error Counts

Figure 10 is intended to summarize the different error categories recorded by the experiment. Error categories must be interpreted using the actual plotting code and matching outputs. In particular, a false positive may arise from a background cluster, a duplicate candidate, a class mismatch, an incorrect temporal match, or a localization error. These causes should not be conflated without further analysis.

VIII. Discussion

The reported experiment demonstrates that the pipeline can process the selected scans and produce candidate regions and evaluation records. It also provides a baseline against which controlled improvements can be tested.

The main limitation is the high number of unmatched proposals. At IoU 0.25, there are 27,214 false positives against 1,028 true positives. At IoU 0.50, the number of false positives rises to 27,991 as the stricter criterion reduces the number of matches. This pattern indicates that proposal generation and localization need substantial improvement.

A post-processing stage that rejects implausible clusters may reduce false positives. Possible signals include cluster size, point count, height, width, length, spatial density, and distance from the sensor. Such filtering must be evaluated on held-out data to avoid overfitting to the subset used for development.

More sophisticated grouping methods may help where DBSCAN merges adjacent objects or fragments sparse objects. However, increasing model complexity before validating the data alignment and evaluation code could obscure the source of errors.

The current results should not be used to claim superiority over learned detectors. Comparisons require identical evaluation rules and appropriate training and validation protocols.

IX. Threats to Validity

A. Ground-Truth Selection

The annotation filter `score == 4` is provisional. If the field represents a different status or confidence convention than assumed, the evaluated ground-truth set may be incorrect. The filter must be verified against official dataset documentation or an authoritative example.

B. Temporal Association

The mapping from scan index to timestamp has not been independently verified. If annotations are associated with different times than the point clouds, the evaluation can report false matches and missed detections that reflect misalignment rather than detector quality.

C. Coordinate Conventions

The physical meaning of the height threshold depends on the coordinate system and units. Center coordinates, box dimensions, heading conventions, and axis order must be confirmed before the geometric calculations can be considered definitive.

D. Approximate Overlap

The current overlap computation uses axis-aligned geometry. It does not fully represent rotated three-dimensional boxes. The reported IoU metrics are therefore approximate and should not be compared directly with results from an oriented 3D benchmark.

E. Greedy Matching

Greedy matching is dependent on candidate ordering and the matching implementation. It may not produce the same assignments as a benchmark evaluator using a different assignment strategy. The evaluation should be checked against a trusted reference implementation.

F. Limited Dataset Coverage

Only 31 scans from Measurement5 were evaluated. This limited scope does not establish robustness across scenes, environments, weather, object classes, or different sensor conditions.

G. Novelty

The use of height filtering, voxel downsampling, and DBSCAN is not itself evidence of algorithmic novelty. A novelty claim would require a documented comparison with prior work and a clearly specified new mechanism or experimentally validated contribution.

X. Recommended Next Steps

The following sequence prioritizes validity before complexity.

1) **Verify annotation semantics.** Confirm the meaning of the score, class, visibility, center, dimension, and heading fields from authoritative documentation.
2) **Verify scan-label alignment.** Establish the correct timestamp and frame association for every point cloud and annotation record.
3) **Validate geometry.** Check coordinate units, axis conventions, box centers, dimension ordering, and heading conventions using visualized examples and known annotations.
4) **Implement oriented 3D IoU.** Test the overlap function on synthetic boxes with known intersections, including rotated boxes and non-overlapping cases.
5) **Audit the matching procedure.** Compare the implementation with a trusted evaluation implementation and verify TP, FP, and FN on hand-checked examples.
6) **Establish a validation split.** Separate development data from evaluation data before performing parameter tuning.
7) **Run controlled parameter sweeps.** Vary voxel size, height threshold, DBSCAN radius, and minimum samples while holding the evaluation protocol fixed.
8) **Investigate proposal filtering.** Test cluster-size and geometry-based filters and record the precision–recall trade-off.
9) **Expand the evaluation.** Evaluate additional subsets and report results by class, distance, object size, and relevant difficulty conditions where the dataset supports these analyses.
10) **Assess novelty separately.** Search for prior work using comparable clustering-based pipelines and define a falsifiable contribution before claiming a new detection mechanism.

XI. Conclusion

This paper presented LUMPI, a preliminary LiDAR object-detection prototype that combines height filtering, voxel downsampling, DBSCAN clustering, cluster-derived candidate boxes, and approximate three-dimensional evaluation.

In the reported 31-scan experiment, the pipeline generated 28,242 candidate proposals and evaluated 1,792 provisionally selected ground-truth boxes. At IoU 0.25, the reported micro-F1 was 0.0685; at IoU 0.50, it was 0.0167. The high false-positive counts and the reduction in matches at the stricter threshold indicate that the baseline requires substantial improvement.

The results are useful as a record of an initial experiment, not as a validated benchmark or a claim of competitive detection. The annotation filter, temporal association, coordinate conventions, and oriented-box evaluation must be validated before stronger conclusions can be drawn.

The next phase should prioritize evaluation correctness, followed by controlled experiments that measure whether each proposed improvement produces a reproducible benefit. Any

claim of scientific novelty should be supported by a precise new method, relevant prior-work comparisons, and falsifiable experiments.

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